

The ordered distribution of electromagnetic waves

according to

- ① frequency
- ② wavelength
- ③ energy

Applications

- ① Radio waves
- ② Microwaves

Poynting Vector $\vec{S}$

A rate of energy travels per $1 \text{ m}^2 \rightarrow \frac{J_s}{\text{m}^2} = \text{W/m}^2$

$$\vec{S} = \vec{E} \times \vec{H}$$

$$\vec{S} = \frac{\vec{E} \times \vec{B}}{\mu_0}$$

Average Poynting vector S_{avg}

$$S_{\text{avg}} = \frac{EH}{2} \rightarrow S_{\text{avg}} = \frac{BHC}{2}$$

$$S = \frac{EB}{2\mu_0} \rightarrow S_{\text{avg}} = \frac{B^2 c}{2\mu_0} \rightarrow S_{\text{avg}} = \frac{\vec{E}^2}{2\mu_0 c}$$

$$E = Bc \text{ or } B = \frac{E}{c}$$

بسم الله الرحمن الرحيم

① An electromagnetic wave travels in free space with $E = 300 \text{ V/m}$. The magnetic field is:

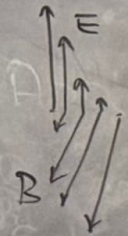
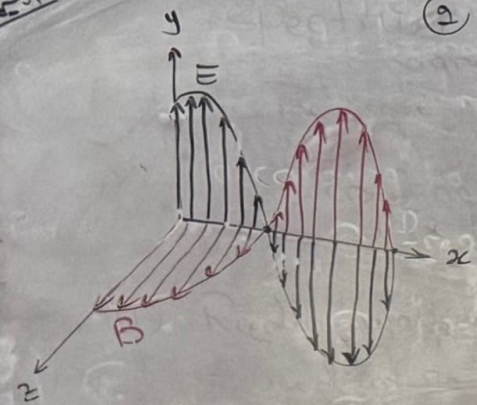
$$B = \frac{E}{c} = \frac{300}{3 \times 10^8} = 100 \times 10^{-8}$$

سوال الیٹرا وائے

① An electromagnetic wave travels in medium ^{where the speed is 80% of speed of light} with $E = 300 \text{ V/m}$. The magnetic field is:

$$B = \frac{E}{c} = \frac{300}{3 \times 10^8 \times 0.8} = 125 \times 10^8 \text{ T}$$

بسم الله الرحمن الرحيم



② An electromagnetic wave having $f = 40 \text{ MHz}$. The wavelength λ is:

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{40 \times 10^6} = \frac{300 \text{ MHz}}{40 \text{ MHz}} = 7.5 \text{ m}$$

An electromagnetic wave having $\lambda = 15 \text{ nm}$. The frequency is:

$$f = \frac{3 \times 10^8}{15 \times 10^{-9}} = 0.2 \times 10^{14} \text{ Hz}$$

بسط المسألة

$$(3) \quad E_y = 600 \sin(10^7 x - \omega t)$$

$$B = \frac{E}{c} = \frac{600}{3 \times 10^8} = 200 \times 10^{-8} \text{ T} \quad (2)$$

$$B_x = 0.6 \sin(3y - \omega t)$$

$$E = Bc = 0.6 \times 3 \times 10^8 = 1.8 \times 10^8 \text{ V/m} \quad (2)$$

(4)

$$\vec{E} = 2\hat{i} + 3\hat{j} - \hat{k}, \quad \vec{B} = 3\hat{i} - 4\hat{j} + 2\hat{k}$$

$$\vec{E} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & -1 \\ 3 & -4 & 2 \end{vmatrix} = \hat{i}(6-4) - \hat{j}(4-3) + \hat{k}(-8-9) \\ = 2\hat{i} - 7\hat{j} - 17\hat{k}$$

$$|\vec{E} \times \vec{B}| = \sqrt{2^2 + 7^2 + 17^2} = 18.5$$

$$\mathcal{E} = \frac{18.5}{4\pi \times 10^7} = 1.47 \times 10^7 \text{ W/m}^2$$

$$\textcircled{5} \vec{E} = 2\hat{i} + 2\hat{j}, \quad \vec{B} = \hat{i} + 6\hat{j}$$

$$\vec{E} = 0.79 \times 10^7 \hat{k} \rightarrow 0.79 \times 10^7 \text{ W/m}^2$$

$$\textcircled{6} \vec{E} = 2\hat{i} + 3\hat{k}, \quad \vec{B} = 3\hat{i} + 6\hat{k}$$

$$\vec{E} = 0.23 \times 10^7 (-\hat{j}) \rightarrow 0.23 \times 10^7 \text{ W/m}^2$$